

BASIC C++ NOTES

1. Introduction to C++

C++ is a general-purpose programming language. It is used for software development, games, operating systems, and competitive programming.

```
#include <iostream>
using namespace std;

int main()
{
    cout << "Hello World";
    return 0;
}
```

2. Variables

A variable is used to store data in memory.

```
int age = 20;
float price = 99.5;
char grade = 'A';
string name = "Saif";
bool isStudent = true;
```

3. Data Types

Common C++ data types are int, float, double, char, string, and bool.

```
int number = 10;
float value = 10.5;
double salary = 25000.50;
char grade = 'A';
string name = "Malik";
bool status = true;
```

4. Input and Output

cout is used to display output and cin is used to take input from the user.

```
int age;

cout << "Enter age: ";
cin >> age;

cout << "Age = " << age;
```

5. Operators

Operators are used to perform operations on values. Common operators are +, -, *, /, %, ==, !=, >, <, &&, and ||.

```
int a = 10;
int b = 5;
cout << a + b;
```

```
cout << a - b;  
cout << a * b;  
cout << a / b;  
cout << a % b;
```

6. Conditional Statements

Conditional statements are used to make decisions in a program.

```
int age = 18;  
  
if(age >= 18)  
{  
    cout << "Eligible";  
}  
else  
{  
    cout << "Not Eligible";  
}
```

7. Switch Statement

The switch statement is used to select one block of code from multiple choices.

```
int choice = 2;  
  
switch(choice)  
{  
    case 1:  
        cout << "One";  
        break;  
  
    case 2:  
        cout << "Two";  
        break;  
  
    default:  
        cout << "Invalid";  
}
```

8. For Loop

A for loop is used to repeat code a fixed number of times.

```
for(int i = 1; i <= 5; i++)  
{  
    cout << i << " ";  
}
```

9. While Loop

A while loop repeats code while a condition is true.

```
int i = 1;  
  
while(i <= 5)
```

```
{
    cout << i << " ";
    i++;
}
```

10. Do-While Loop

A do-while loop executes the code at least once.

```
int i = 1;

do
{
    cout << i << " ";
    i++;
}
while(i <= 5);
```

11. Arrays

An array stores multiple values of the same data type.

```
int numbers[5] = {10, 20, 30, 40, 50};

cout << numbers[0];
```

12. Functions

A function is a block of code that performs a specific task.

```
int add(int a, int b)
{
    return a + b;
}

int main()
{
    cout << add(10, 20);
    return 0;
}
```

13. Basic C++ Program Structure

A C++ program commonly contains header files, namespace, the main function, statements, and a return statement.

```
#include <iostream>
using namespace std;

int main()
{
    cout << "Welcome to C++";

    return 0;
}
```